Question:
What is the order of the element 5 in U_8?
Answer Choices:
A. 3
B. 4
C. 1
D. 5
E. 9
F. 2
G. 8
H. 0
I. 6
J. 7

Reasoning + Answer:

To find the order of the element 5 in the group $ \mathbb{ Z }_8 $ (the cyclic group of integers modulo 8), we need to determine the smallest positive integer $ k $ such that :

$$
k \cdot 5 \equiv 0 \pmod{ 8 }
$$

This is the definition of the order of an element in a cyclic group . The order of an element $ a $ in $ \mathbb{ Z }_n $ is given by :

$$
\text{ ord }(a) = \frac{n }{ \gcd (n , a)}
$$

Here , $ n = 8 $, $ a = 5 $

So :

$$
\gcd (8 , 5) = 1
$$

Thus ,

$$
\text{ ord }(5) = \frac{ 8 }{ 1 } = 8
$$

Therefore , the order of 5 in $ \mathbb{ Z }_8 $ is 8 .

<answer> G </answer>